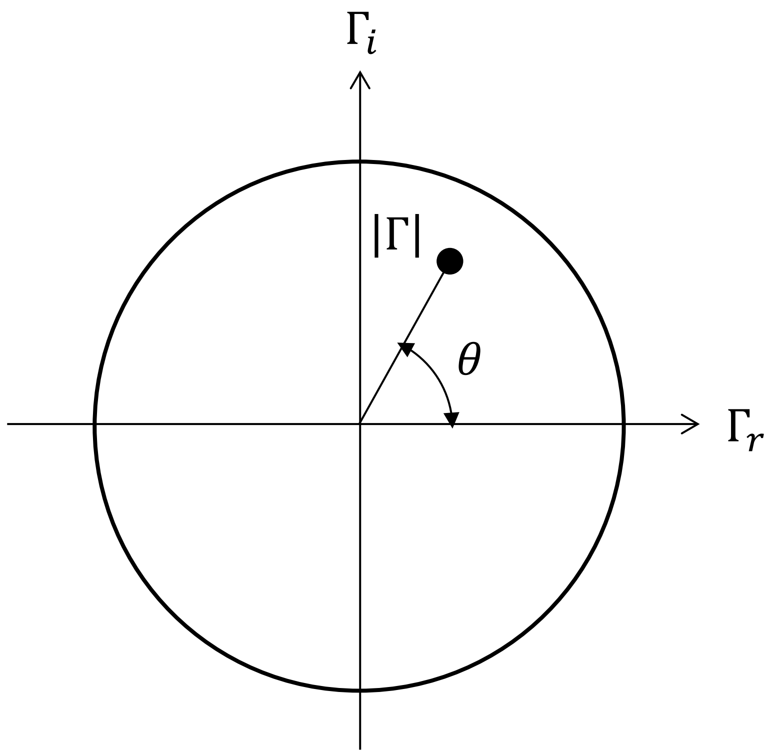
Smith chart

The Smith chart was invented in 1939 by P. H. Smith at Bell labs, as a graphical tool to help solving transmission line problems.



It's a graphical representation of the reflection coefficient is polar coordinates

$$\Gamma = |\Gamma| e^{j\theta} = \Gamma_r + j\Gamma_i$$

Since $0 \leq |\Gamma| \leq 1$ the Smith chart is contained within a circle of radius 1

Normalized impedance

Recall the normalized impedance: $z_L = \frac{Z_L}{Z_0}$:

$$\Gamma = \frac{z_L - 1}{z_L + 1} = |\Gamma| e^{j\theta} \qquad (17)$$

Solving for z_L:

$$z_L = \frac{1 + |\Gamma| e^{j\theta}}{1 - |\Gamma| e^{j\theta}} \qquad (29)$$

Separating the real and imaginary parts:

$$z_L = r_L + jx_L \qquad \Gamma = \Gamma_r + j\Gamma_i$$

$$r_L + jx_L = \frac{1 + \Gamma_r + j\Gamma_i}{1 - \Gamma_r - j\Gamma_i}$$

Multiplying numerator and denominator by $1 - \Gamma_r + j\Gamma_i$:

$$r_L = \frac{1 - \Gamma_r{}^2 - \Gamma_i{}^2}{(1 - \Gamma_r)^2 + \Gamma_i{}^2} \quad (29a); \qquad x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i{}^2} \quad (29b)$$

Parametric equations

Eqns. (29) can be rearranged as parametric equations, where r_L and x_L (the impedance) are expressed in terms of Γ_r and Γ_i (the horizontal and vertical axis on the Smith chart):

$$\boxed{\begin{aligned} &\left(\Gamma_r - \frac{r_L}{1+r_L}\right)^2 + \Gamma_i^{\ 2} = \left(\frac{1}{1+r_L}\right)^2 \qquad (31a) \\ &(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2 \qquad (31b) \end{aligned}}$$

Notice that Eqns. (31) are of the same form as:

$$(x - x_0)^2 + (y - y_0)^2 = a^2$$

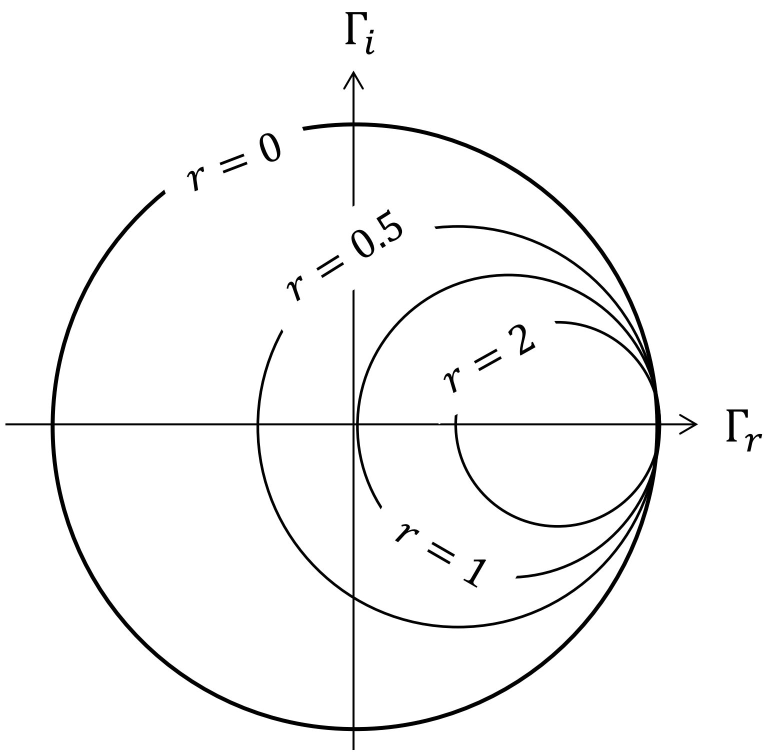
Which describes a circle of radius a, centred at the point (x_0, y_0)

(31a) is the **resistance circle**

(31b) is the **reactance circle**

Resistance circles

$$\left(\Gamma_r - \frac{r_L}{1+r_L}\right)^2 + \Gamma_i^{\ 2} = \left(\frac{1}{1+r_L}\right)^2 \quad \Rightarrow x_0 = \frac{r_L}{1+r_L};\ \ y_0 = 0;\ \ \ a = \frac{1}{1+r_L}$$

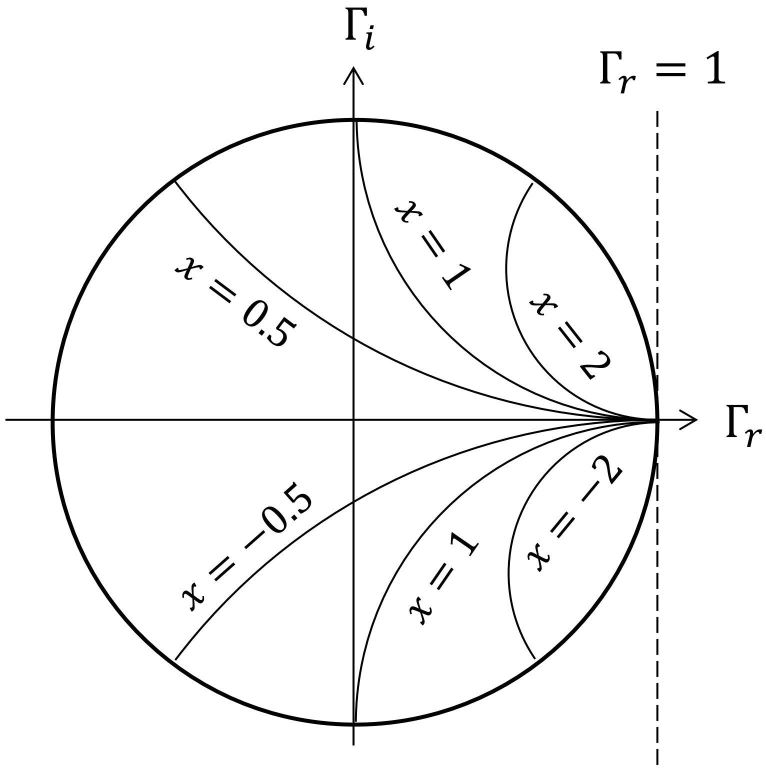


All resistance circles touch the point (1,0) to their right, and are centred along the positive real axis.
The radius is 0 for $r = \infty$ (open circuit) and 1 for $r = 0$

Note: $r = 0$ is not necessarily a short circuit! It could be a pure inductor or capacitor.
However, there is only one way to make $r = \infty$

Reactance circles

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2 \Rightarrow x_0 = 1; \;\; y_0 = \frac{1}{x_L}; \quad a = \frac{1}{|x_L|}$$

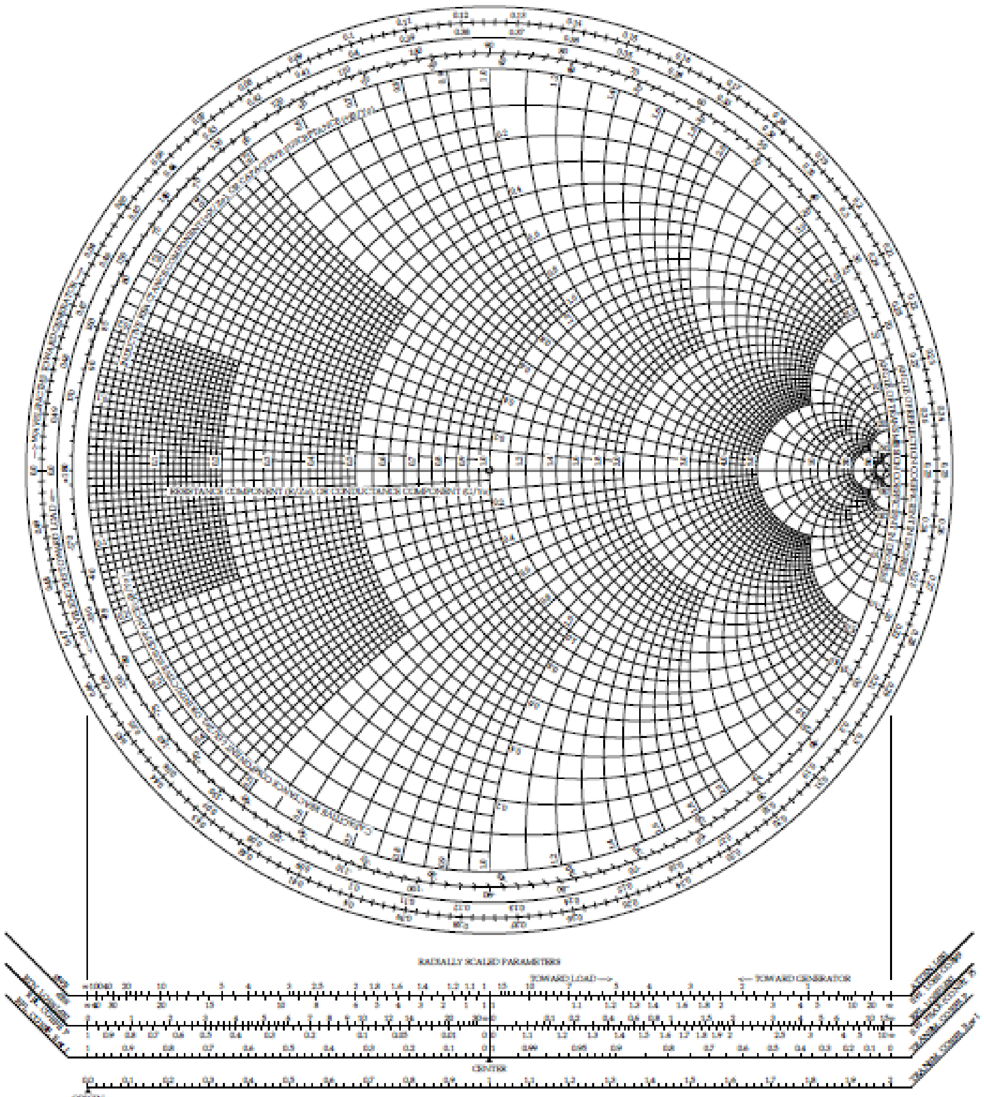


All reactance circles are centred along the vertical line $\Gamma_r = 1$ and touch the $(1{,}0)$ point.

The radius is equal to the vertical shift of the centre.

There's no sense in drawing the reactance circles outside the circle of unit radius, since there is no physical value of r compatible with that.

Full Smith chart



Resistance and reactance circles always cross orthogonally to each other

Reflection coefficient on the Smith chart

Given a (normalized) impedance $z_L = r_L + jx_L$:

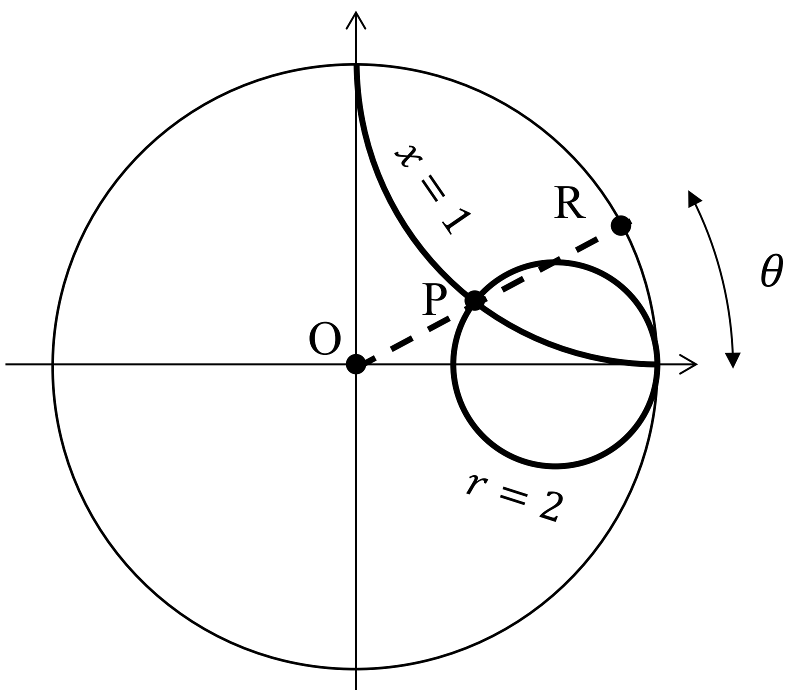
- The reflection coefficient Γ is found immediately by marking the intersection of the r_L and x_L (point P).

- $|\Gamma|$ is the length of the segment, relative to the radius of the chart: $|\Gamma| = \text{OP/OR}$.

- θ is found by reading the angle scale around the circle labelled "angle of reflection coefficient"

Reflection coefficient on the Smith chart

Example: $z_L = 2 + j1$



$$\Gamma = 0.45e^{j26.6^o}$$

Input impedance

Recall Eq. (24) for the input impedance of a transmission line, at a distance l from the load:

$$Z_{in} = Z_0 \frac{1 + \Gamma\, e^{-j2\beta l}}{1 - \Gamma\, e^{-j2\beta l}} \qquad (24)$$

Normalizing it by the characteristic impedance, and substituting $\Gamma = |\Gamma| e^{j\theta}$:

$$\boxed{z_{in} = \frac{Z_{in}}{Z_0} = \frac{1 + |\Gamma|\, e^{j(\theta - 2\beta l)}}{1 - |\Gamma|\, e^{j(\theta - 2\beta l)}} \qquad (32)}$$

Eq. (32) is identical to Eq. (29), after decreasing the phase of the reflection coefficient by an angle $2\beta l$.

On the Smith chart, **the input impedance is found by rotating clockwise by an angle $2\boldsymbol{\beta l}$ on the circle of radius $|\boldsymbol{\Gamma}|$, starting from the point P determined by $\boldsymbol{z_L}$.**

Rotations around the Smith chart

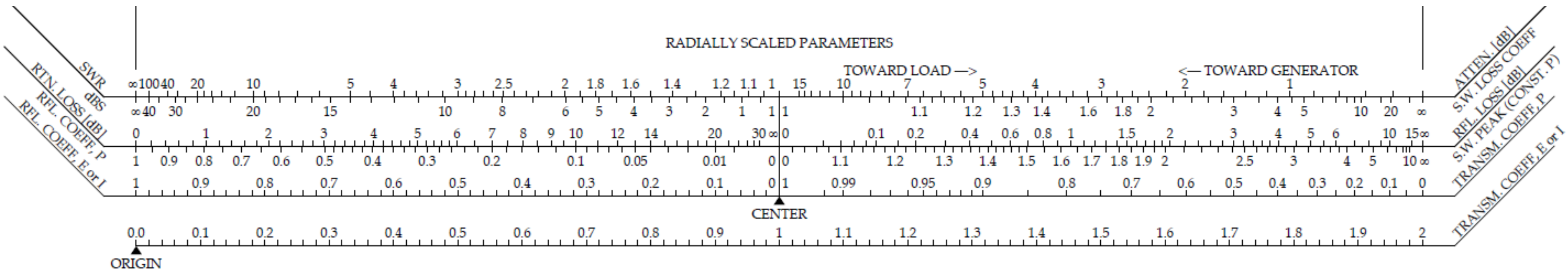
The Smith chart has four scales around its periphery. From the outside inwards:

1. Wavelengths towards the generator: this is a relative scale, used to perform clockwise rotations of angle $2\beta l$ to find the input impedance at distance l "backwards", from the load towards the generator.
2. Wavelengths towards the load: this is a relative scale, used to perform anti-clockwise rotations of angle $2\beta l$ to find the input impedance at distance l "forwards", from the generator towards the load.
3. Angle of reflection coefficient: used to find θ (see example before).
4. Angle of transmission coefficient. We won't use it.

Recall that $\beta = \frac{2\pi}{\lambda}$. When $l = \frac{\lambda}{2}$, the rotation $2\beta l$ covers an angle 2π, i.e. you go a full circle around the chart. We saw this before – a half-wavelength line acts as if it weren't there!

Radially scaled parameters

Under the Smith chart you find several horizontal scales that can be used to find the parameters of the system:

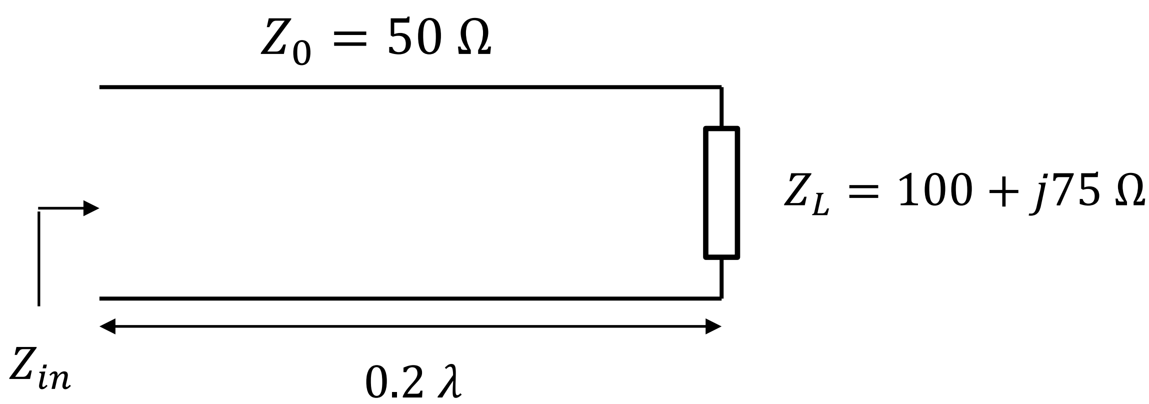


With a ruler, measure the length of the segment OP, where P is the point representing the normalized impedance on the Smith chart, then use that length to find out:

- Standing wave ratio SWR
- Reflection coefficient, E or I (the "P" refers to power, $|\Gamma|^2$)
- Return Loss
- Etc…

Example

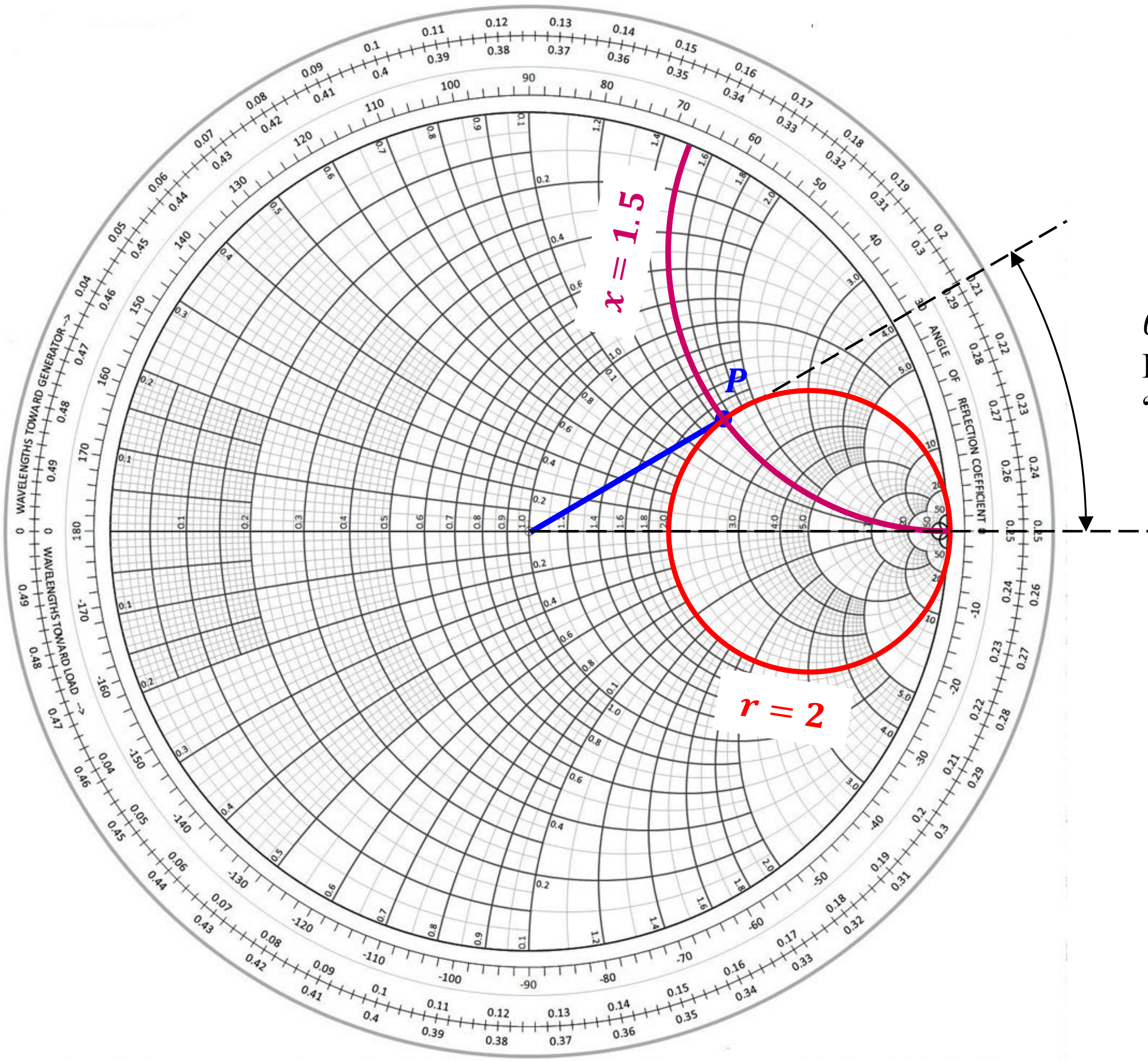
Consider the following circuit:



Normalized load impedance:

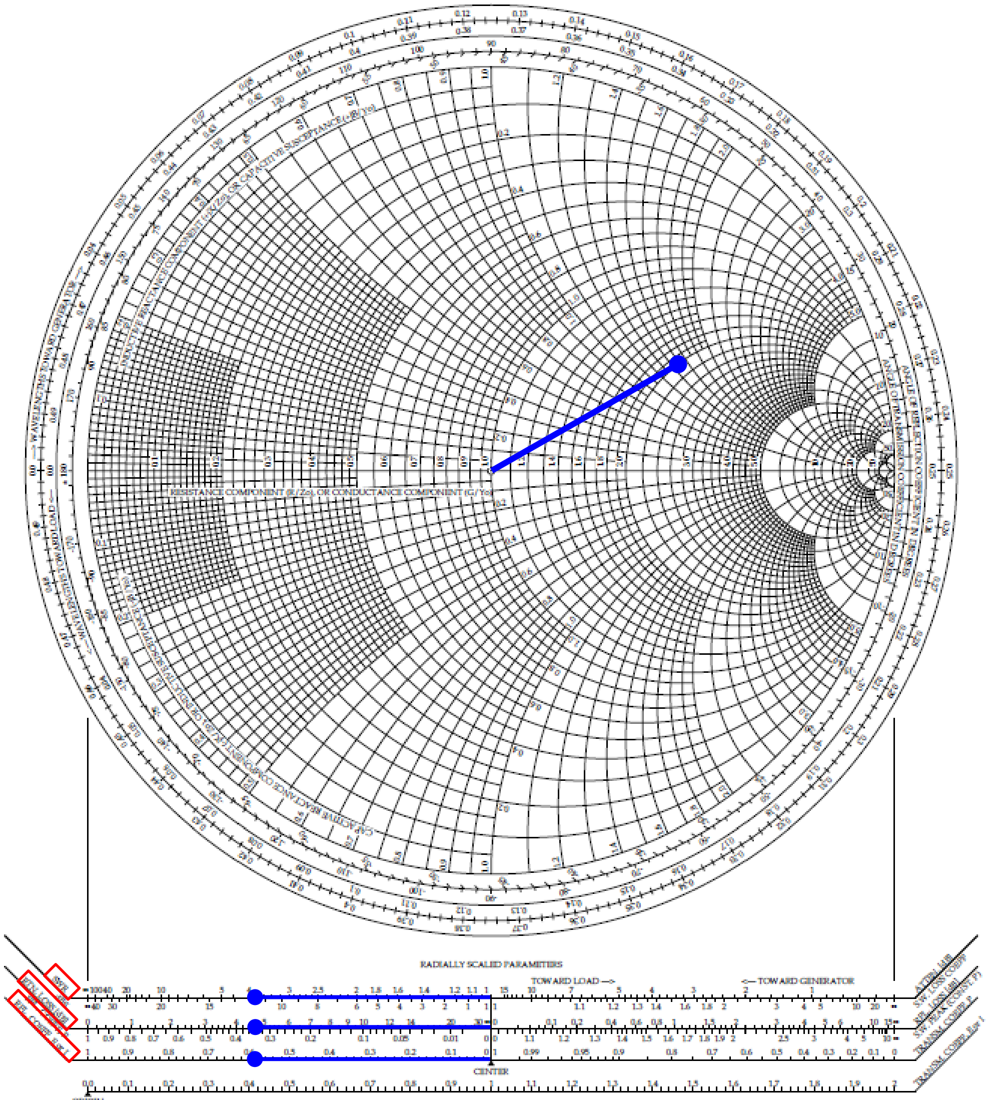
$$z_L = 2 + j1.5$$

At the load



$\theta = 30^o$
From the scale
“angle of reflection coefficient”

Parameters

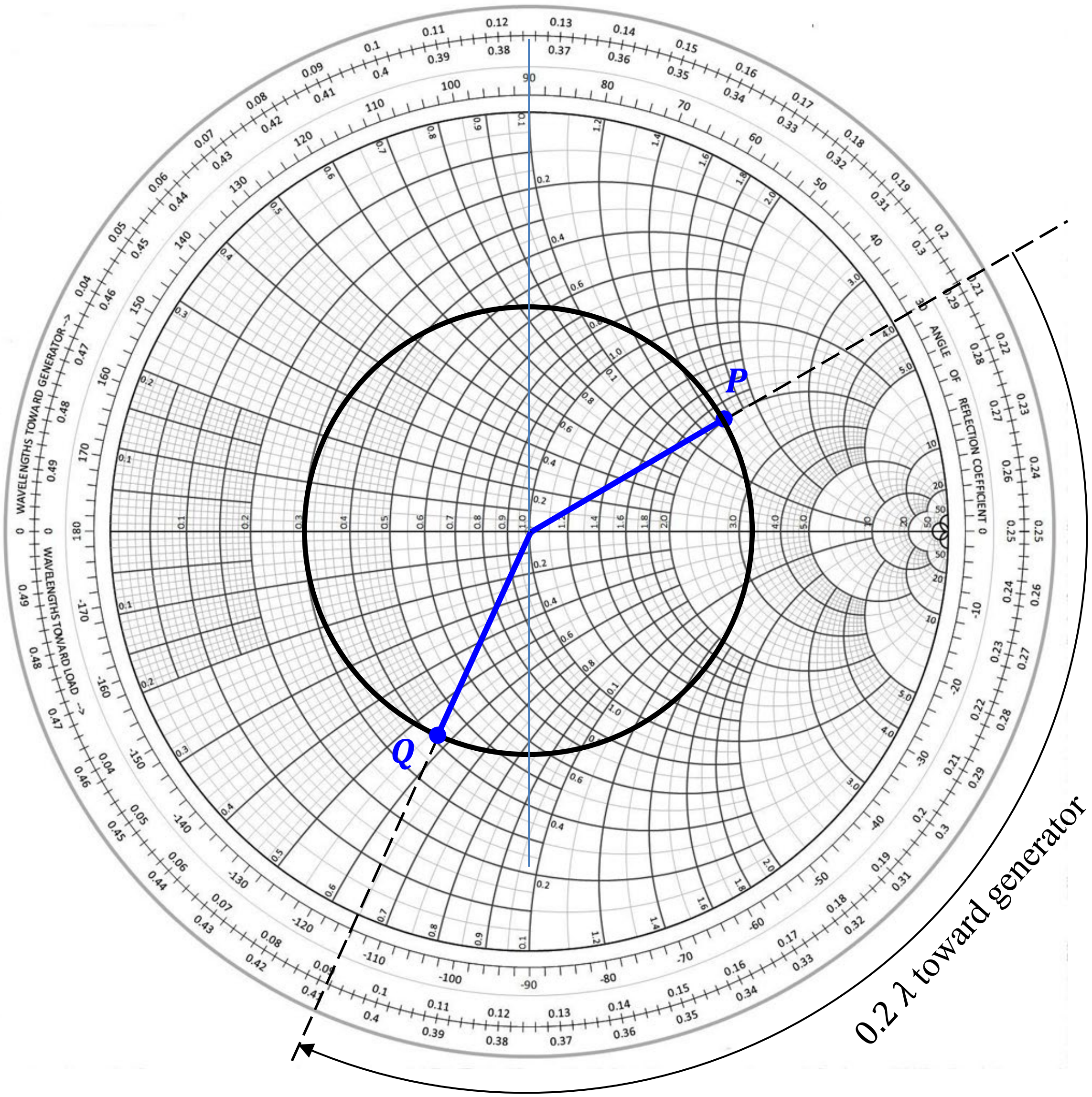


SWR = 3.8

RTN loss = 4.7 dB

$|\Gamma| = 0.59$

At the input



$z_{in} = 0.42 - j0.57$
$\theta = -114^o$

All other parameters stay the same because they only depend on $|\Gamma|$

The black circle centred at the origin, $|\Gamma| = \text{const.}$ is sometimes called "S circle", because S (= VSWR) is constant along it.

Impedance → admittance conversion

Recall the effect of introducing a $\lambda/4$ line (page 46):

$$Z_{in} = \frac{{Z_0}^2}{Z_L}$$

In normalized form:

$$z_{in} = \frac{1}{z_L} = y_L$$

Where y_L is the normalized admittance of the load.

On the Smith chart:

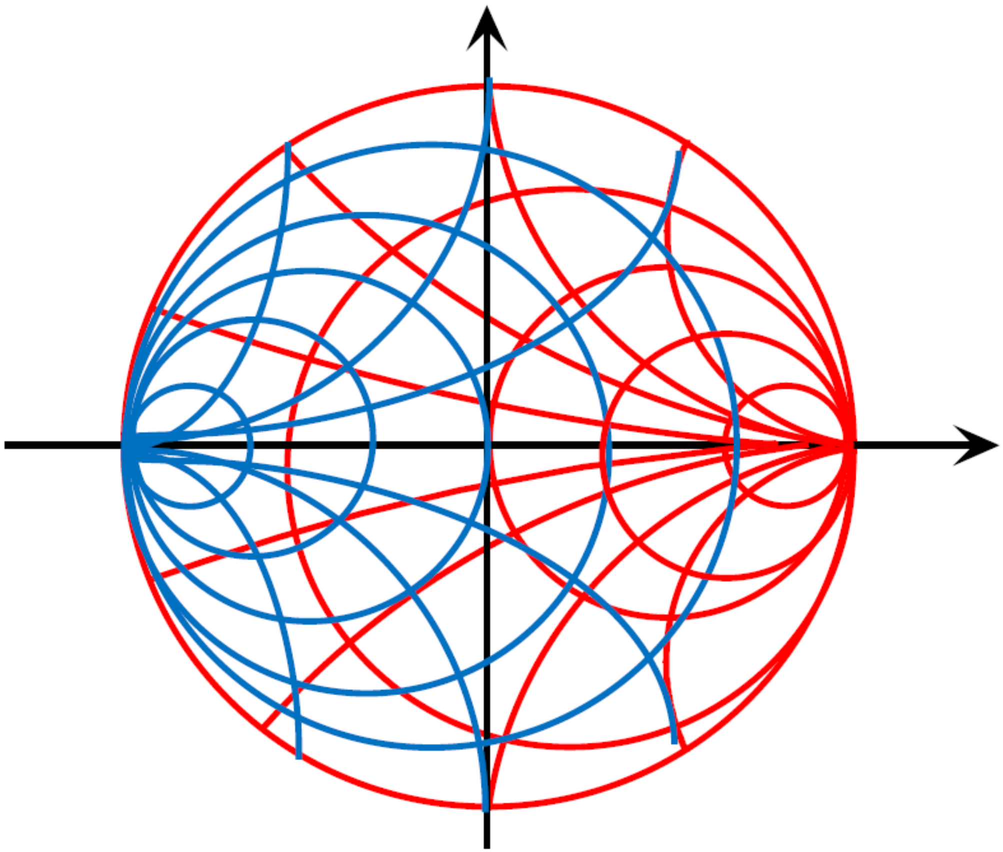
A rotation of 180^o on the Smith chart converts a normalized impedance into the corresponding normalized admittance.

Sometimes called "flipping the chart".

After the 180^o rotation you can interpret the values on Smith chart, **as is**, as representing admittances instead of impedances.

Combined Impedance - admittance chart

You can find combined Smith charts that plot both impedance and admittance, with different colours.

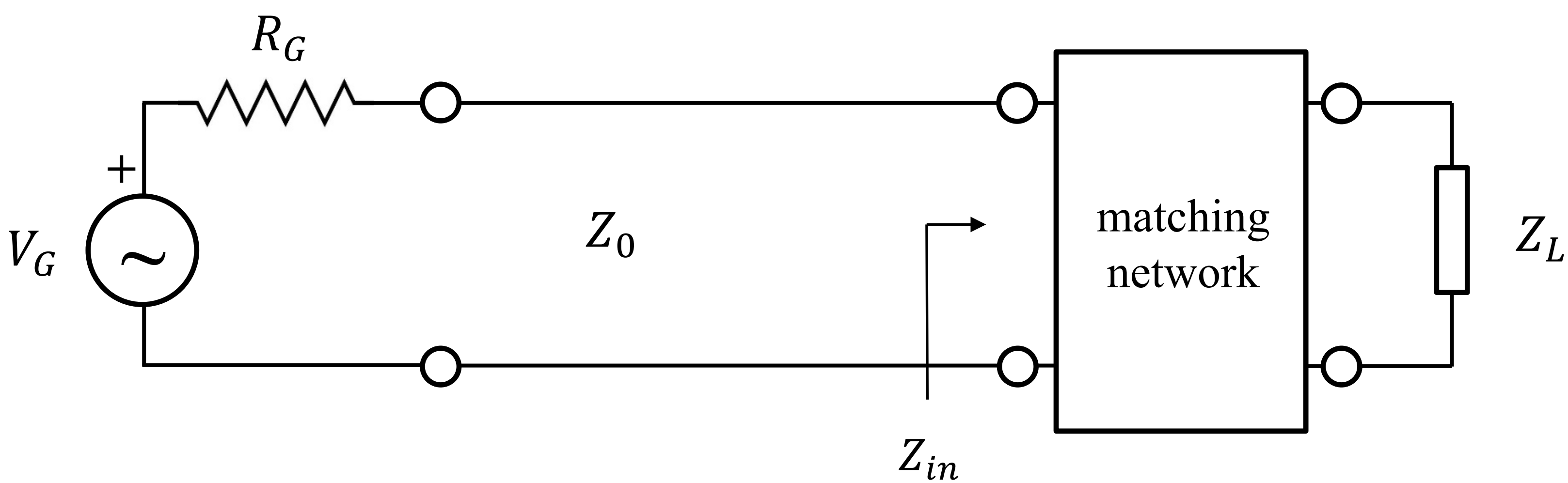


Red circles: impedance
Blue circles: admittance

Here you can read directly *z* and *y* without "flipping the chart". However, this chart becomes a bit messy, so we won't use it.

Impedance matching networks

An impedance matching network is a circuit that presents an input impedance $Z_{in} = Z_0$ even when connected to a load $Z_L \neq Z_0$. Therefore, the network "transforms" the unmatched load into something that appears matched.



We will normally assume $R_G = Z_0$

Matching on the Smith chart

On the Smith chart, a matched load correspond to a point at the origin, $\Gamma = 0$.

So we need to find a “path” to go from our unmatched impedance, point P, to the origin, point O.

In terms of normalized impedance, we want to achieve $z_{in} = 1 + j0$.

Recall that, by simply adding a piece of transmission line, we can rotate around the chart on the circle $|\Gamma| = \text{const}$.

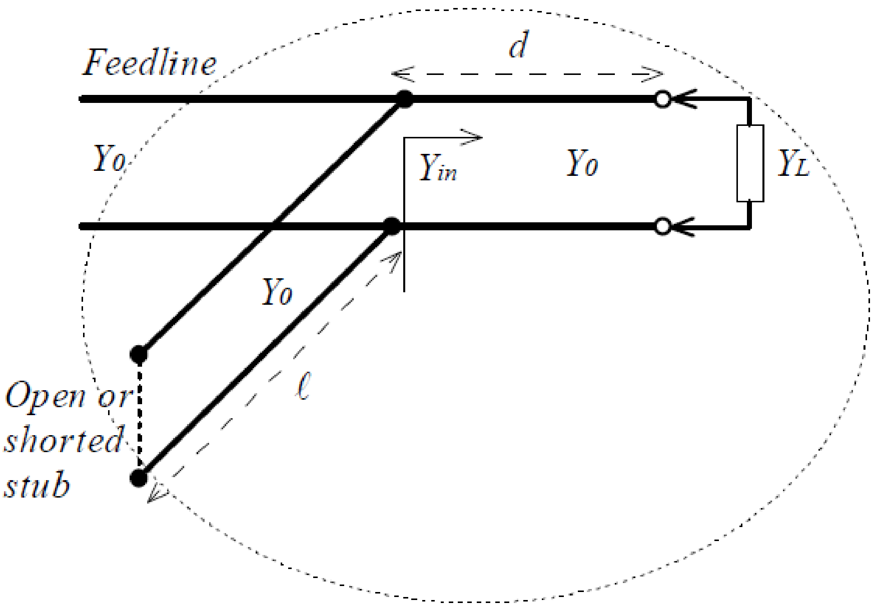
Basic idea:

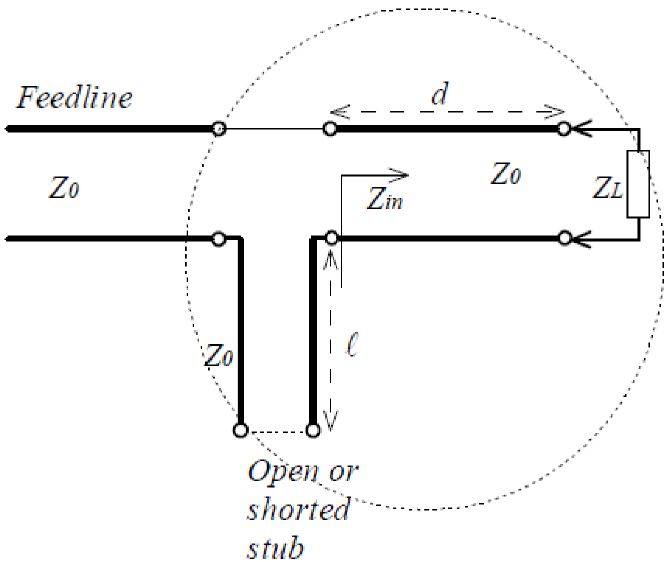
Step 1: Add a length of line that brings you to intersect the $r = 1$ circle. Now you have matched the real part of the impedance, but the reactance might still be nonzero.

Step 2: Add a reactance that cancels the residual reactance after matching the resistive component. This can be done by adding something in series or in parallel to the line.

Single-stub matching

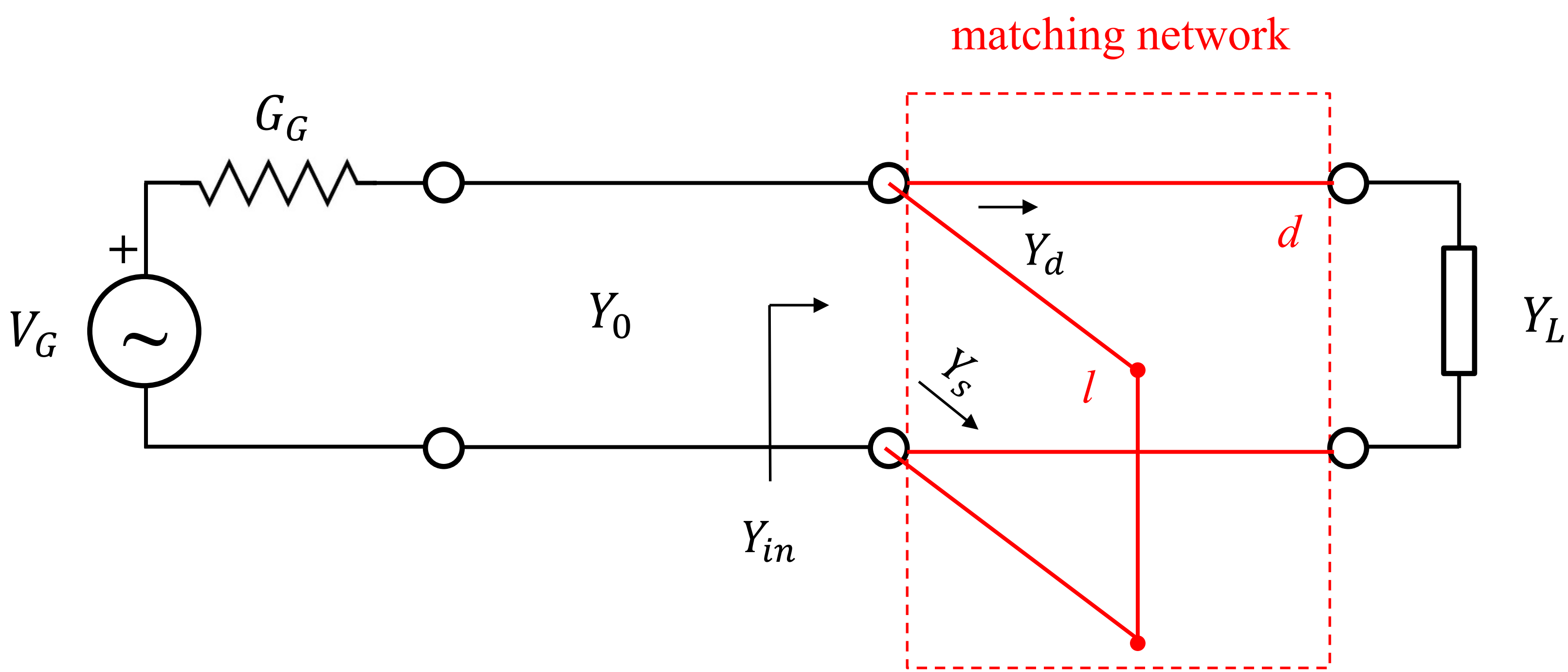
A piece of transmission line of length *l* (called "stub"), terminated by a short or an open circuit, is added in parallel or in series to the line (called "feedline"), at a distance *d* away from the load





Short-circuited stub in parallel

It is often easier to insert stubs in parallel, both in coaxial cables and in microstrip. Also, a short circuit is usually more "perfect" than an open circuit, where fringing fields and other imperfections can be substantial at high frequencies.



Short-circuited stub in parallel

When inserting components in parallel, it's easier to use the admittances, since they simply add:

$$Y_{in} = Y_d + Y_s$$

In normalized units:

$$y_L = \frac{1}{z_L} = g_L + jb_L$$

Calling $y_d = g_d + jb_d$ the normalised admittance looking at the input of the feedline, the steps to match the load are:

Step 1: calculate *d* such that $g_d = 1$

Step 2: calculate *l* such that $b_s = -b_d$

with these steps we obtain a matched load $y_{in} = 1$

Note: since the stub is terminated by a short circuit, the admittance you see upstream from it is always imaginary, i.e. $g_s = 0$ (think about it…)

Matching procedure

For **shunt short-circuited stub**:

1. Calculate the normalized load impedance $z_L = Z_L/Z_0$ and locate it on the Smith chart. This is point P_1.
2. Draw the S circle through P_1.
3. "Flip the chart", by drawing a line from P, through O, until it intersects the S circle on the opposite side. This is point P_2 and represents the normalized admittance of the load. From now on, the chart will indicate admittances.
4. Extend the line P_1O P_2 to the periphery of the Smith chart, and mark the point P_2' where the line crosses the "Wavelengths toward generator" scale. Write down the value of λ_2 at P_2'. Remember: the WTG scale is relative. You should think of P_2' as the origin of that scale from now on, because that's where the load admittance is.

Matching procedure

5. Mark the points P_3 and P_4 where the S circles intersects the circle $g = 1$. Note again: the circle $g = 1$ is what "used to be" $r = 1$ but now represents admittances because you've flipped the chart once. Notice that $y_{3,4} = 1 \pm jb$, i.e. the susceptances differ only by the sign.
6. Extend the lines OP_3 and OP_4 to the periphery of the chart, and mark the points P_3' and P_4' on the WTG scale. Write down the values of the corresponding λ_3 and λ_4.
7. Calculate $d_3 = \lambda_3 - \lambda_2$ and $d_4 = \lambda_4 - \lambda_2$. d_a and d_b (relative to the wavelength λ) are two possible position where $g_d = 1$. You have now matched the real part of the admittance (**Step 1**).
8. Now you can attach a shunt stub at point d_a or d_b to cancel out the imaginary part of the admittance. Since we are considering a short-circuited stub, we start from point P_{sc} which, <u>for admittances</u>, is on the right edge of the chart ($y = \infty$). Turning clockwise, calculate the lengths l_3 and l_4 that take you from P_{sc} to P_3", where $y = -jb_3$ and P_4", where $y = -jb_4$. $P_{3,4}$" are on the outer edge of the chart (**Step 2**).

Remarks on matching

The single-stub matching technique we have discussed (as well as most other methods) is **frequency-dependent**. It relies upon a stub being of a certain length with respect to the wavelength. At a different frequency, λ is different so the stub has the wrong length!

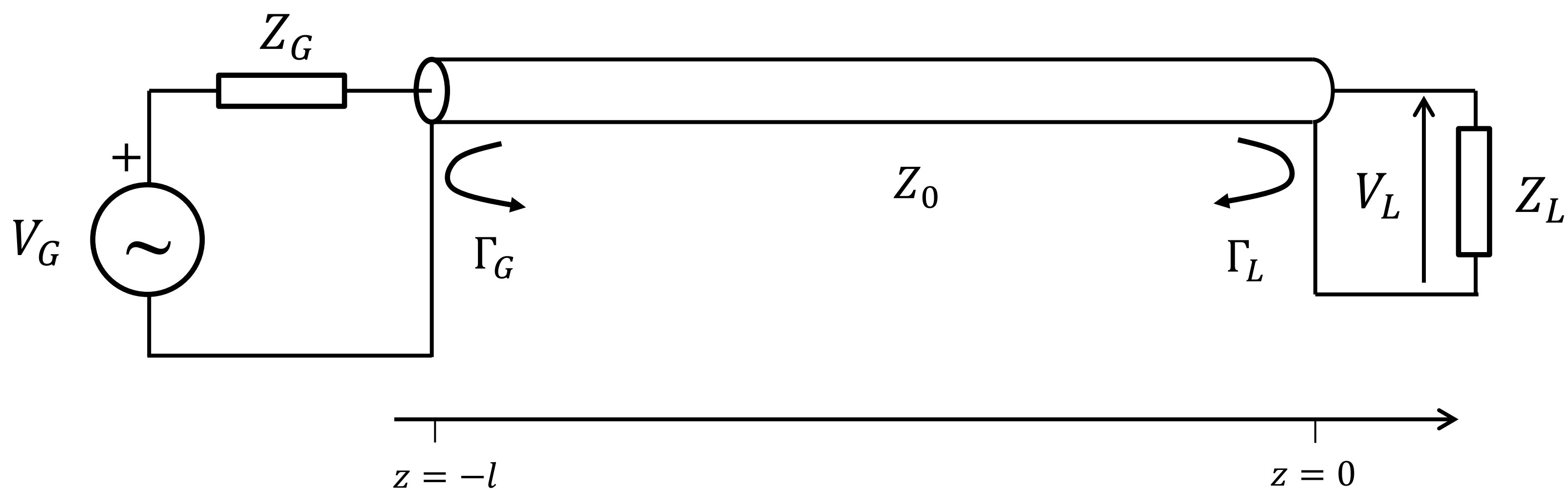
For the same reason, the analysis we did was all based upon phasors, i.e. we considered steady-state sinusoidal excitation at a single frequency. What happens if we apply a step? A big mess. A step function contains a broad spectrum of frequencies, and the response of the circuit isn't matched at all.

Stub-based matching presents a matched impedance to the generator at a specific frequency, but it does so by causing reflections between the stub and the load.

Amplitude of incident waves

In the preceding lectures, we have seen how to treat the propagation of wave with voltage amplitude V_0^+.

Consider the general circuit below:



How do you calculate V_0^+?

Amplitude of incident waves

In the case of a **transient in the time domain** (like when using time-domain reflectometry):

$$V_0^+ = V_G \frac{Z_0}{Z_0 + Z_G}$$

This is because, at the instant it turns on, the generator only "sees" the impedance of the cable.

In the case of **steady-steady harmonic signals** (which we describe with phasors), it's not so simple!!

There is still an incident and a reflected wave if there is some mismatch, but finding the correct value of V_0^+ requires accounting for all reflections between generator, line and load.

Amplitude of incident waves

Using Eqns. (18a) and (24):

$$V(z) = V_0^+\left(e^{-j\beta z} + \Gamma_L\, e^{+j\beta z}\right); \qquad Z_{in} = Z_0 \frac{1 + \Gamma_L\, e^{-j2\beta l}}{1 - \Gamma_L\, e^{-j2\beta l}}$$

At the generator ($z = -l$) the total voltage must satisfy the voltage division rule:

$$V(-l) = V_G \frac{Z_{in}}{Z_{in} + Z_G} = V_0^+\left(e^{j\beta l} + \Gamma_L\, e^{-j\beta l}\right)$$

From this we extract the value of V_0^+:

$$V_0^+ = V_G \frac{Z_{in}}{Z_{in} + Z_G} \frac{1}{\left(e^{j\beta l} + \Gamma_L\, e^{-j\beta l}\right)} \qquad (33)$$

Or, using Eq. (24):

$$V_0^+ = V_G \frac{Z_0}{Z_0 + Z_G} \frac{e^{-j\beta l}}{\left(1 - \Gamma_G \Gamma_L\, e^{-j2\beta l}\right)} \qquad (34)$$

For steady-state harmonic signals, V_0^+ is not a trivial quantity!